

Search for Ultra-Relativistic Magnetic Monopoles with LHAASO-KM2A

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Ultra-relativistic magnetic monopoles traversing the atmosphere are expected to initiate extensive air showers characterized by a muon deficit and a compact lateral profile. Using 1438.81 live days of data from the LHAASO-KM2A array, we present a search for these cosmic particles within the Lorentz factor range of 10^5 to 10^7 . A machine-learning classifier utilizing lateral shower shape observables from the electromagnetic detectors is deployed to separate the signal from the hadronic background. No statistical excess above the background expectation is observed, and upper limits at the 90% confidence level are established on the isotropic monopole flux. The resulting upper limit is $9.7 \times 10^{-17} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ at $\gamma = 10^5$ and reaches $8.1 \times 10^{-19} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ at $\gamma = 10^7$ which improves upon the astrophysical Parker bound by over three orders of magnitude. This work establishes constraints on the monopole flux within a previously unconstrained velocity window.

Introduction.—Magnetic monopoles (MMs) are hypothetical particles originally postulated by Dirac [1] in 1931 to restore electromagnetic symmetry and explain the quantization of electric charge. They were subsequently shown to be mandatory topological defects arising from spontaneous symmetry breaking in Grand Unified Theories [2, 3]. The discovery of MMs would have important implications for fundamental physics, shedding light on charge quantization mechanisms, phase transitions in the early Universe—such as cosmic inflation and the associated cosmological monopole prob-

lem [4, 5]—and the ultimate unification of forces at inaccessible energy scales. While MMs are generally expected to be massive and stable, their exact phenomenology remains subject to substantial theoretical uncertainty due to their strong-coupling nature and unknown mass scale. Experimentally, MMs carrying a single unit of magnetic charge interact with ordinary matter with an electromagnetic coupling strength significantly larger than that of an electron. This key feature provides a clear experimental signature through distinctive energy-loss mechanisms [6], which have been widely utilized to search for

MMs.

Over the past few decades, numerous experiments have been conducted to search for MMs, primarily categorized into accelerator-based and cosmic-ray-based experiments. Collider searches are sensitive to MMs with masses ranging from the GeV to TeV scales, probing production cross sections down to the femtobarn level [7, 8]. Conversely, cosmic-ray searches can probe much larger mass scales and are sensitive to extremely rare fluxes down to the order of $1.0 \times 10^{-21} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ [9]. There exists also the astrophysical Parker bound [10] based on the survival of the Milky Way’s magnetic field, which severely restricts the possible density of monopoles in the universe and predicts that the MM flux shall be smaller than $1.0 \times 10^{-15} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ for $\beta < 10^{-3}$, becoming less constraining in highly relativistic velocity regimes [11, 12].

Currently, the leading limits on the cosmic monopole flux are established by different detectors depending on the velocity regime. The MACRO experiment [13] sets the most competitive constraints in the non-relativistic regime, and extends into the relativistic region alongside IceCube [14] for velocities up to $\beta \approx 0.995$. In the ultra-relativistic domain, the Pierre Auger Observatory [9] dominates at extreme energy scales with Lorentz factors $\gamma > 10^7$. Critically, a clear gap in sensitivity persists in the regime where γ falls below 10^7 , where the sensitivity to MMs flux is not compellingly limited by the detection technologies used in these cosmic-ray observatories.

In this Letter, we present the first search for such cosmic MMs using the Large High Altitude Air Shower Observatory square-kilometer array (LHAASO-KM2A) [15]. With its large fiducial volume and continuous exposure, the array is well-positioned to fill this phase-space gap. Its dense, co-located arrays of electromagnetic detectors (EDs) and muon detectors (MDs) provide strong background rejection and fine spatial granularity, making it well-suited to search for MMs with Lorentz factors between 10^5 and 10^7 . In this ultra-relativistic window, penetrating MMs are expected to initiate extensive air showers through continuous interactions. For this study, an intermediate mass scale of 10^5 GeV is chosen as a reference benchmark. According to the energy loss formulas of monopoles [6, 16], bremsstrahlung radiation scales inversely with the square of the projectile mass and is heavily suppressed compared to pair production and photonuclear effects at this scale. Consequently, while this work utilizes this specific baseline, the established simulation framework provides a valid reference for other mass regimes where bremsstrahlung radiation remains similarly negligible.

Detector and Simulation.—The LHAASO observatory, situated at an altitude of 4410 m above sea level, is dedicated to high-energy cosmic-ray and gamma-ray astronomy covering a wide energy range from 10 TeV

to 100 PeV. This analysis utilizes data recorded by KM2A, which comprises 5216 EDs and 1188 underground MDs [17]. The dense layout of the ED array provides high-granularity sampling of the electromagnetic component of air showers, enabling precise reconstruction of both the core position and the lateral distribution profile. Concurrently, the extensive coverage of the MD array allows for an efficient measurement of the secondary muon content for each shower. The combination of these two subsystems offers powerful discrimination to distinguish hypothetical ultra-relativistic MMs—which are characterized by muon deficit and a compact lateral profile—from the hadronic cosmic-ray background. The dataset analyzed in this Letter was accumulated between July 2021 and July 2025 for a total of 1438.81 live days, corresponding to a stable operation period with the fully deployed array configuration [18].

Monte Carlo (MC) simulations are performed using CORSIKA [19] integrated with CONEX [20, 21] to model the extensive air showers. Standard hadronic cosmic-ray backgrounds are generated using QGSJET-II-04 [22] for high-energy regimes and GHEISHA [23] for low-energy components. For ultra-relativistic MMs with $M = 10^5$ GeV, the continuous ionization energy loss is supplemented by stochastic processes [6, 16]. Importantly, the total energy loss in this high-energy regime is dominated by pair production and photonuclear interactions, while bremsstrahlung radiation is suppressed by the large mass of monopole [16]. Signal samples are simulated at three discrete Lorentz factors: $\gamma = 10^5$, 10^6 , and 10^7 . Secondary particles reaching the ground are processed through G4KM2A [24], the official LHAASO detector simulation based on GEANT4 [25], which accurately accounts for the realistic geometry, material budgets, and electronic trigger responses of the KM2A array.

Analysis Methodology.—Extensive air showers induced by ultra-relativistic MMs develop primarily through electromagnetic pathways. The absence of strong hadronic interactions will result in a deficit of secondary muons at the LHAASO observation level compared to standard cosmic-ray events. Furthermore, owing to their stable characteristics, monopoles continuously deposit energy through sequential interactions with atmospheric nuclei along their trajectories [9]. This energy drain induces multiple overlapping sub-showers, which reconstruct as a single, highly compact shower profile. Crucially, unlike standard cosmic-ray cascades that undergo rapid atmospheric attenuation after reaching their developmental maxima, monopole showers exhibit a persistent growth tendency sustained by these continuous secondary injections [16]. Consequently, monopole-induced cascades exhibit a younger shower age (s_{age}) and a denser lateral distribution. This distinctive penetrating depth implies that for large zenith angles (θ_{rec}), where backgrounds are heavily attenuated by the extended atmospheric grammar, the survival and detection probabilities of isotropic

MMs remain extremely high, providing a topological signature for experimental discrimination in Fig. 5.

Owing to the dominance of electromagnetic energy loss, monopole-induced showers behave similarly to those initiated by high-energy gamma rays. Therefore, the presence of gamma-ray showers presents a contribution to cosmic background that can limit the sensitivity of such rare-signal search. To suppress this background while minimizing the associated selection efficiency loss, the search region excludes the galactic plane within $|b| < 10^\circ$ as well as all known cataloged Galactic and extragalactic gamma-ray sources [26]. This precise spatial masking is enabled by the excellent angular resolution of the KM2A array, which is determined using Crab Nebula data to be better than 0.2° for energies above 100 TeV [27] for cosmic rays.

The remaining diffuse gamma-ray flux from unmasked regions is expected to be negligible. For extragalactic components, long-distance propagation induces strong attenuation via pair production on cosmic background radiation fields [28, 29]. Consequently, no direct observation of this diffuse flux exists at these energies, leaving only conservative upper limits. Notably, even if a minor residual flux survives these physical suppressions, it is inherently accounted for within the data-driven background estimation to be detailed below, ensuring no bias in the final search window.

To separate the rare magnetic monopole signals from the overwhelming hadronic cosmic-ray background, the signal region (SR) is established using two complementary observables: the reconstructed shower energy (E_{rec}) and the muon-to-electromagnetic ratio ($R_{\mu e}$). The former, E_{rec} , is reconstructed via an XGBoost model [30] trained specifically on simulated monopole samples. Utilizing a linear-fit baseline, the model takes the reconstructed zenith angle and the lateral particle density at 50 m from the shower core (ρ_{50}) as inputs, where ρ_{50} is obtained from an NKG-function fit [31, 32] to the shower front. These two observables establish a log-linear scaling between the shower size and the reconstructed energy, aligning the energy scales of the background and signal events to enhance the signal purity for specific kinematic configurations. The latter, $R_{\mu e}$, parameterizes the secondary muon content and is defined as $R_{\mu e} = \log_{10}[(N_\mu + 10^{-4})/N_{\text{EM}}]$, where N_μ and N_{EM} denote the particle counts within 200 m for MDs and 400 m for EDs from the shower core [27], respectively. Because standard hadronic showers possess a higher muon yield, this ratio provides a useful handle to suppress the background. To maximize the experimental sensitivity, the specific selection thresholds for both observables are optimized independently for each discrete Lorentz factor $\gamma = 10^5$, 10^6 , and 10^7 , defining the customized SR boundaries summarized in Table I.

Events in the zero-muon limit ($R_{\mu e} < -5$) are excluded from the SR. For $\gamma = 10^5$, this is due to the

TABLE I: Optimized selection ranges for $\log_{10}(E_{\text{rec}}/10^8/\text{GeV})$ and $R_{\mu e}$, defining the SR for each discrete Lorentz factor γ .

γ	$\log_{10}(E_{\text{rec}}/10^8/\text{GeV})$	$R_{\mu e}$
10^5	[1.95, 2.10]	$[-5.0, -2.9]$
10^6	[2.85, 3.10]	$[-5.0, -2.6]$
10^7	[3.75, 4.20]	$[-5.0, -2.2]$

much worsened signal-over-background ratio and the incompatibility to apply the data-driven background modeling templates extracted from the muon-populated control regions. For the higher Lorentz factors ($\gamma \geq 10^6$), the enormous energy scale of the showers makes zero-muon fluctuations physically precluded. Therefore, the exclusion of zero-muon events have no impact on the high-energy search while ensuring a consistent treatment across all investigated kinematic points.

The distributions of the primary selection observables are presented in Fig. 4. As illustrated, the $R_{\mu e}$ thresholds define an extreme muon-poor window, suppressing the background survival fraction to below 5×10^{-6} . Based on simulation, the residual cosmic-ray background surviving within this region is heavily dominated by primary protons weighted by the Gaisser composition model [33], which account for over 99% of the remaining hadronic component. However, due to this massive rejection power, the surviving MC sample size inside the SR is depleted to near-zero events, resulting in direct simulation-driven background modeling highly unstable and unfeasible. Furthermore, in this muon-poor regime near the SR boundary, predictions from alternative hadronic interaction models exhibit substantial discrepancies, limiting their reliability as a simulation baseline [24, 34].

To circumvent these limitations, a novel data-driven approach [35, 36] is developed to evaluate the hadronic background, removing dependencies on both the hadronic generators and primary mass composition assumptions. The method is grounded in an intrinsic feature of hadronic air showers within the control region (CR): at a fixed E_{rec} range, the shower lateral profiles stabilize asymptotically with decreasing $R_{\mu e}$. This asymptotic convergence is verified across six shape observables (see Fig. 5 and definition in End Matter), which show excellent convergence in the low- $R_{\mu e}$ region. Based on this stability, the background shape inside the SR is parameterized using an empirical template extracted from a background-dominated data sample adjacent to the SR boundary with $-2.9 < R_{\mu e} < -2.85$.

To further exploit the topological differences between signal and background showers, a Multi-Layer Perceptron (MLP) [37] is utilized for multivariate classification. The network takes six observables as inputs to differentiate the simulated monopole signals from the data-driven

background template. Training is performed using a 5-fold cross-validation strategy to ensure stability and mitigate overfitting, yielding a mean validation AUC of 0.951. The output score of the trained MLP is subsequently employed as the final discriminant in the likelihood fit to establish the flux upper limits.

Systematics and Statistical Framework.—Systematic uncertainties are evaluated separately for the data-driven background and the simulated signal components. For the background, the normalization is treated as a free-floating parameter in the global fit, leaving shape variations as the primary systematic source. For the signal, the hadronic model uncertainties encompass both the overall selection efficiency and the differential distribution shapes.

The background shape uncertainty is evaluated by constructing a symmetric systematic envelope from the maximum bin-by-bin deviations among alternative control templates extracted across adjacent $R_{\mu e}$ cut slices (see Fig. 5). A Gaussian smoothing algorithm is subsequently applied to mitigate local statistical fluctuations, defining the continuous shape variations.

Signal simulation uncertainties primarily originate from the choice of the hadronic interaction model, which dictates the simulated shower development. To quantify this model dependence, the selection efficiencies of the combined E_{rec} and $R_{\mu e}$ cuts are cross-evaluated using alternative generators, including EPOS-LHC v3400 [38] and SIBYLL 2.3d [39]. The maximum spread among these hadronic models is assigned as the normalization uncertainty on the signal efficiency, yielding 4.1%, 9.3%, and 1.8% for $\gamma = 10^5$, 10^6 , and 10^7 , respectively. Similarly, systematic shape variations in the signal MLP output distribution are parameterized using the maximum bin-by-bin discrepancy between these alternative models.

The statistical inference is performed using a binned profile likelihood fit across the final MLP score distribution. The parameterized likelihood function $\mathcal{L}(\mu, \mu_{\text{bkg}}, \theta)$ is formulated as:

$$\mathcal{L}(\mu, \mu_{\text{bkg}}, \theta) = \prod_{i=1}^{N_{\text{bins}}} \text{Pois}(n_i | \nu_i) \times \prod_k \mathcal{LN}(\theta_{\text{sig},k} | 1, \sigma_k) \times \prod_m \mathcal{G}(\theta_{m,s} | 0, 1), \quad (1)$$

where n_i is the observed event count in the i -th bin, and the expected yield $\nu_i = \mu \cdot s_i(\theta_{\text{sig},k}, \theta_{\text{sig},s}) + \mu_{\text{bkg}} \cdot b_i(\theta_{\text{bkg},s})$ is a superposition of the signal and background templates with indices k, s denoting efficiency and shape. Here, μ represents the signal strength, θ denotes the vector of nuisance parameters, and μ_{bkg} is an unconstrained free-floating multiplier for the background normalization. The background normalization factor, μ_{bkg} , is introduced to compensate for residual yield biases predicted by the

data-driven extrapolation and its post-fit values consistently fall within [1.1, 1.3] in the global fit. Multiplicative efficiency errors are constrained by log-normal terms $\mathcal{LN}$, while the Gaussian terms $\mathcal{G}$ govern the template shape morphing ($\theta_{m,s}$, where $m \in \{\text{sig}, \text{bkg}\}$).

To test the compatibility of the data with the background-only hypothesis, we employ the profile likelihood ratio test statistic $q_\mu = -2 \ln[\mathcal{L}(\mu, \hat{\theta}(\mu)) / \mathcal{L}(\hat{\mu}, \hat{\theta})]$, where $\hat{\theta}(\mu)$ denotes the conditional maximum-likelihood estimators of the nuisance parameters for a specified μ , and $\hat{\mu}$ and $\hat{\theta}$ represent the unconditional global maximums [40].

Results.—No significant excess over the background expectation is observed in the final MLP score distributions. To validate the background-only hypothesis, the post-fit distributions for discrete Lorentz factors $\log_{10}(\gamma) = 5, 6$, and 7 are presented in Figs. 1(a)–1(c). The experimental data exhibit excellent compatibility with the background predictions within the total post-fit $\pm 1\sigma$ uncertainty bands.

TABLE II: Expected ($\pm 1\sigma$) and observed 90% C.L. upper limits on the isotropic magnetic monopole flux.

$\log_{10}(\gamma)$	Expected Limit [$10^{-19} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$]	Observed Limit
5	$1201.5^{+585.1}_{-379.0}$	973.2
6	$19.0^{+20.8}_{-7.9}$	11.6
7	$8.2^{+0.2}_{-0.2}$	8.1

Consequently, upper limits on the isotropic magnetic monopole flux are established at the 90% confidence level (C.L.) using the CL_s method [43]. The resulting expected ($\pm 1\sigma$) and observed limits for a reference mass of $M = 10^5$ GeV are summarized in Table II. Figure 2 presents the LHAASO-KM2A upper limits in comparison with the astrophysical Parker bound [10] and existing constraints from Pierre Auger [9], RICE [41], and ANITA-II [42].

Conclusion.—We have performed the first search for ultra-relativistic MMs using the fully deployed LHAASO-KM2A array. Focusing on a reference monopole mass of $M = 10^5$ GeV, this analysis probes a critical kinematic window spanning Lorentz factors $10^5 \leq \gamma \leq 10^7$, which remains valid for other mass scales where pair production and photonuclear interactions dominate. By leveraging a strict secondary muon number selection, a data-driven background estimation and an MLP multivariate discriminant, the overwhelming hadronic cosmic-ray background is strongly suppressed, yielding an expectation of less than one background event in the signal regions for two higher Lorentz points. Upon unblinding, no statistically significant signal excess is observed comparing with background-only hypothesis.

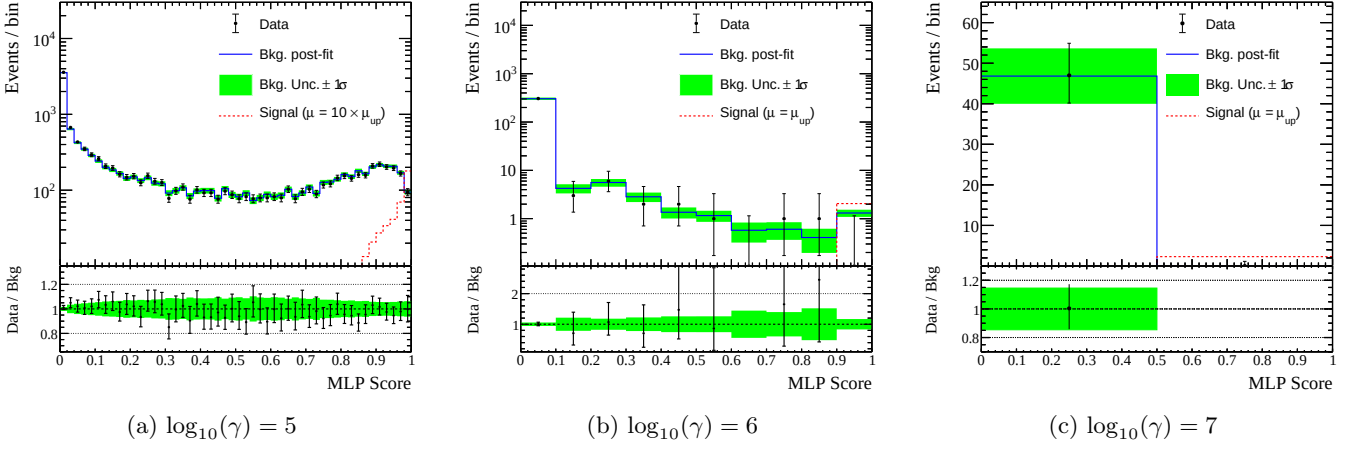


FIG. 1: Post-fit MLP score distributions for $\log_{10}(\gamma) =$ (a) 5, (b) 6, and (c) 7. Data points are compared with background-only fit expectation histograms; shaded bands indicate the total $\pm 1\sigma$ uncertainties.

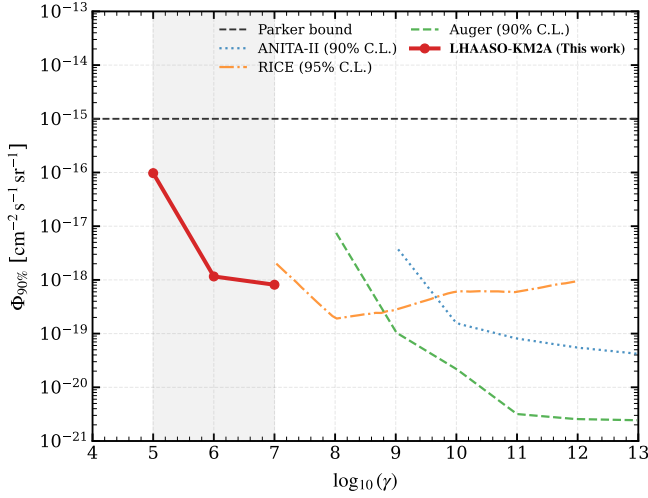


FIG. 2: The 90% C.L. upper limits on the magnetic monopole flux from LHAASO-KM2A (red solid line), compared with the astrophysical Parker bound [10] and existing constraints from Pierre Auger [9], RICE [41], and ANITA-II [42].

The resulting 90% C.L. upper limits set a stringent experimental boundary on the isotropic cosmic monopole flux. At the highest boost factor investigated ($\gamma = 10^7$), the derived limit of $8.1 \times 10^{-19} \text{ cm}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$ directly constrains models of astrophysical monopole production, matching and extending the experimental reach of ultra-high-energy cosmic-ray observatories such as Pierre Auger and RICE. At lower Lorentz factors ($\gamma = 10^5, 10^6$), where radio and giant surface arrays lose detection efficiency due to high cascade-energy thresholds, the LHAASO-KM2A constraints successfully fill a long-standing phase-space gap. Overall, the derived limits remain more than one to three orders of magni-

tude below the Parker bound across the entire investigated ultra-relativistic window, advancing the experimental constraints on such exotic particles.

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MLP Inputs Definition.—To separate the hypothetical monopole signal from cosmic-ray background, MLP classifier is trained using six shower observables. Let q_i denote the total integrated charge (in photoelectrons, PE) registered by the i -th active ED unit, and $r_i = \sqrt{(x_i - x_{\text{core}})^2 + (y_i - y_{\text{core}})^2}$ represent its lateral distance to the reconstructed shower core. The definitions for these variables are as follows:

(1) **Reconstructed Zenith Angle** (θ_{rec}): Reconstructed via a timing fit to the relative arrival times of secondary particles across the active array. Large zenith angles provide a deep atmospheric filter that suppresses standard hadronic backgrounds.

(2) **Shower Age** (s_{age}): A shape parameter indicates the steepness of the lateral distribution function, extracted by fitting the recorded lateral particle densities with a modified NKG function [31, 32].

(3) **Outer Core Density** (ρ_{40}): The average charge per hit registered in the outer core region beyond a radial boundary of 40 m, defined as $\rho_{40} = \sum_{i \in \mathbb{I}_{40}} q_i / N_{40}$, where $\mathbb{I}_{40} = \{i \mid r_i > 40 \text{ m}\}$ and $N_{40} = |\mathbb{I}_{40}|$.

(4) **Pincness**: The shower is segmented into concentric radial bins b with a width of 10 m, adapted from reference [44]. For bins containing $N_b \geq 2$ hits, Pincness = $\frac{1}{N_{\text{valid}}} \sum_b \sum_{i \in \text{bin } b} [(\zeta_i - \bar{\zeta}_b) / \sigma_b]^2$, where $\zeta_i = \log_{10}(q_i + 1)$, $\bar{\zeta}_b$ and σ_b are the sample mean and standard deviation of ζ within bin b , and N_{valid} is the cumulative number of hits inside all valid bins.

(5) **Compactness**: The ratio of the total hit multiplicity to the maximum localized charge recorded at a large lateral radius ($r > 45 \text{ m}$), defined as Compactness = $N_{\text{hit}} / \max_{i \in \mathbb{I}_{45}} (q_i)$ where $\mathbb{I}_{45} = \{i \mid r_i > 45 \text{ m}\}$, adapted from reference [45]. Monopole events exhibit higher compactness values due to the absence of peripheral hadronic sub-cores.

(6) **Mean Radius** (R_{mean}): The charge-weighted mean lateral distance of all triggered ED units, parameterizing the global lateral dispersion of the energy deposition: $R_{\text{mean}} = \sum_{i=1}^{N_{\text{hit}}} q_i r_i / \sum_{i=1}^{N_{\text{hit}}} q_i$, where N_{hit} is the total number of hit detectors.

Background Modeling Validation.—The data-driven background estimation assumes shape invariance of the multi-variable shower parameters across adjacent $R_{\mu e}$ sidebands. Fig. 3 illustrates this behavior through an example of scanning of s_{age} with variation and shift most obvious under different $R_{\mu e}$ slices.

The core physical mechanism governing this conver-

gence is rooted in the high-energy fluctuations of cosmic-ray air showers. As mapped by the s_{age} distribution and its variation in Fig. 3, the longitudinal shower development exhibits large structural mutations in the high-muon regimes ($R_{\mu e} \geq -1.5$, yellow dotted lines). However, as $R_{\mu e}$ continuously decreases, the secondary hadronic interaction is depleted, forcing the atmospheric cascades into the pure electromagnetic fluctuation limit. Below the critical boundary of $R_{\mu e} \leq -2.5$, the longitudinal profiles asymptotically freeze into an invariant configuration (dark solid lines). This stabilization is formalized in the inset panels: the mean profile $\langle s_{\text{age}} \rangle$ flattens into a rigid plateau, with its nominal residual variation $\Delta \langle s_{\text{age}} \rangle$ bounded within 0.01.

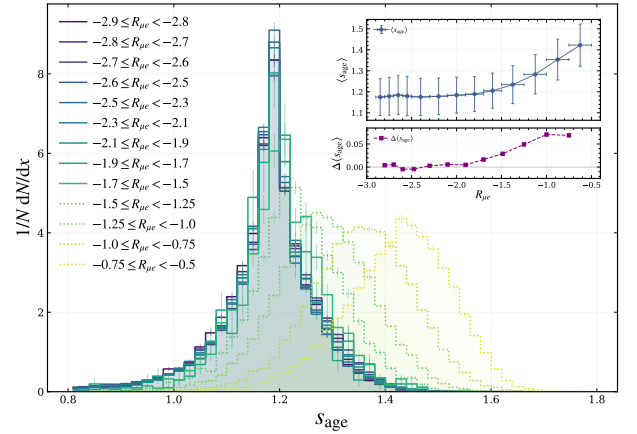


FIG. 3: Normalized shower age (s_{age}) distribution scanned across different $R_{\mu e}$ slices.

This shape stability inside the deep control region also holds for the other five observables shown in Fig. 5. For other two phase points $\gamma = 10^6, 10^7$, observable distributions are shown in Supplemental Material. By capitalizing on this asymptotic convergence, the empirical template extracted from the sideband provides an unbiased representation of the background in the signal region. The small variations across adjacent sidebands are conservatively treated as a shape systematic uncertainty in the profile likelihood fit. In contrast, the simulated monopole signal shows a completely different distribution across all variables, confirming the strong discrimination power of these engineered parameters.

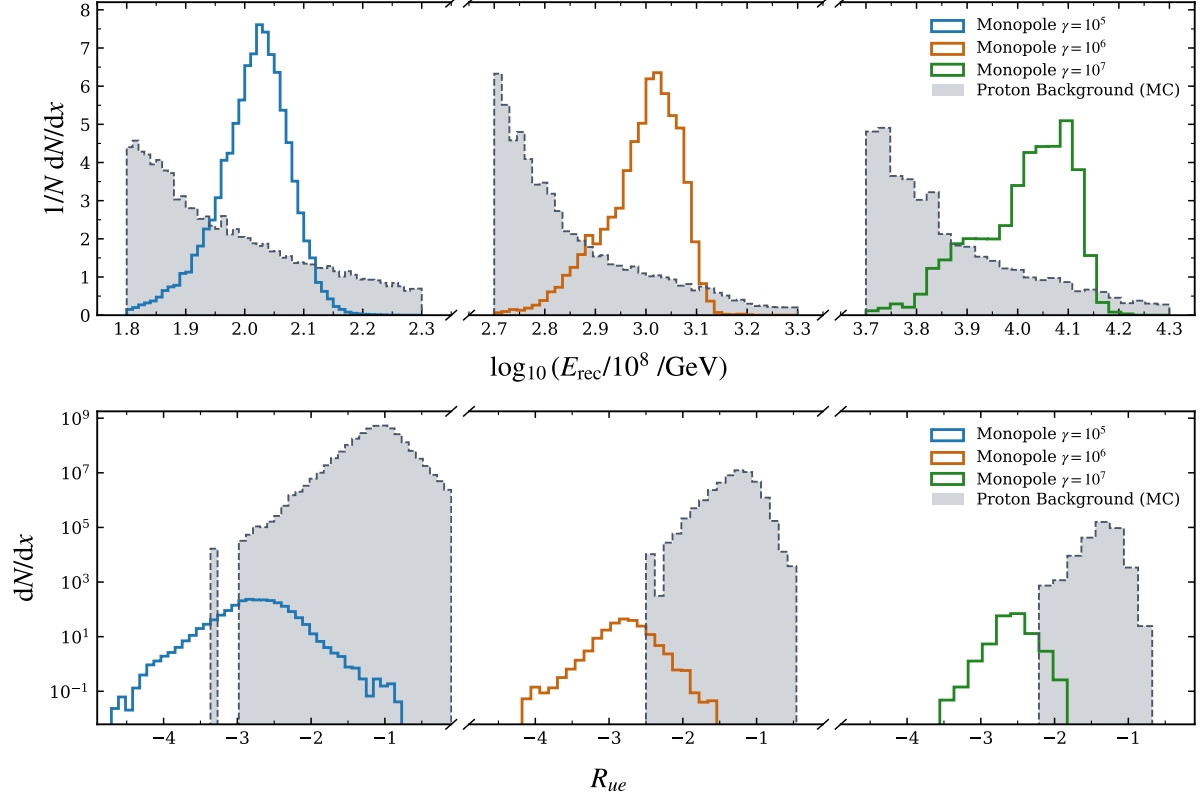


FIG. 4: Distributions of the primary selection variables for the simulated magnetic monopole signals at $\gamma = 10^5$, 10^6 , and 10^7 compared against the simulated proton with Gaisser composition weight: (a) reconstructed energy $\log_{10}(E_{\text{rec}}/10^8 \text{ GeV})$ and (b) muon-to-electromagnetic ratio $R_{\mu e}$.

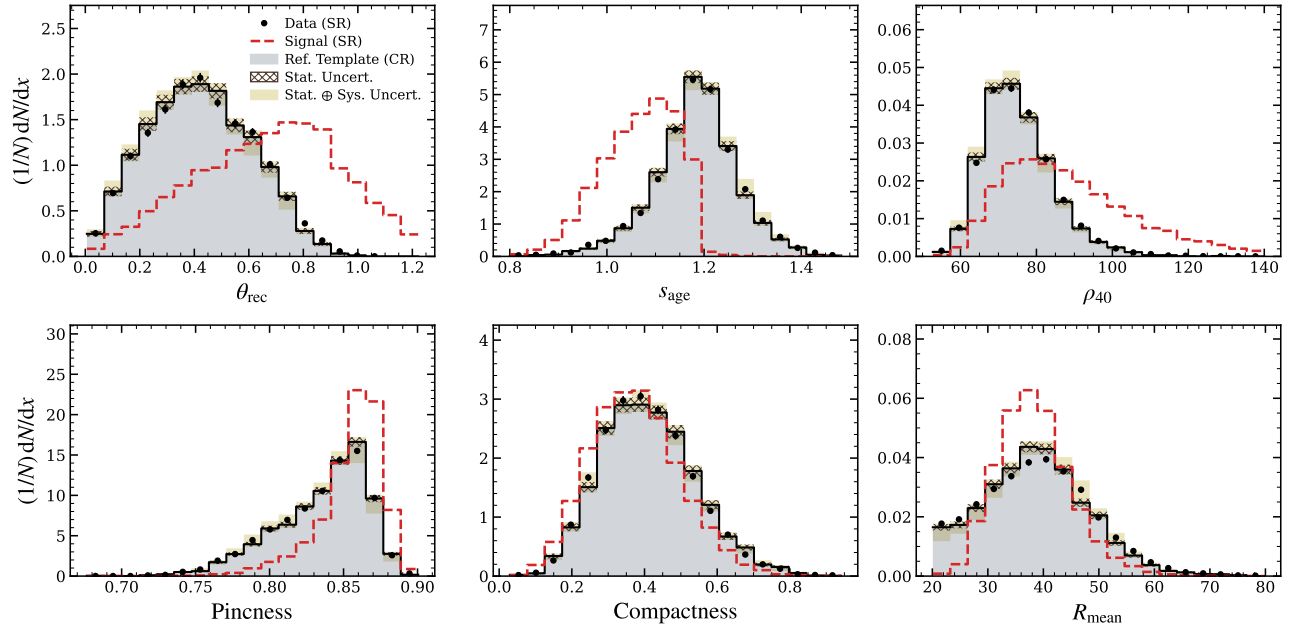


FIG. 5: Validation of the shower shape invariance across the six MLP input observables for the $\gamma = 10^5$ analysis. The reference template (black step) is compared against adjacent control slices (incorporated in yellow shadow) and the simulated monopole signal topology (red dashed line). The data in SR (black dots) show excellent consistency with templates extracted from CR.